Inequality Exercises: Constraints, Equality, and Vector Geometry

N-20220811

The central skill in an inequality problem is finding the correct representation of the expression. Fractions can be rewritten as a Cauchy–Schwarz inner product; a radical can encode a Euclidean distance; a quadratic constraint defines an ellipsoid or sphere. The standard methods and equality conditions follow Hardy, Littlewood and Pólya's *Inequalities* and Strang's MIT 18.06 linear algebra material.[1]

§1.1 Fraction inequalities and their domains

For positive denominators u, v, Cauchy–Schwarz gives

$$\frac{a^2}{u} + \frac{b^2}{v} \geq \frac{(a+b)^2}{u+v}.$$

To see equality, write

$$\frac{a^2}{u} + \frac{b^2}{v} - \frac{(a+b)^2}{u+v} = \frac{(av-bu)^2}{uv(u+v)}.$$

Thus equality occurs exactly when $a/u = b/v$.

The original expression

$$F(x,y) = \frac{4}{4-x^2} + \frac{9}{9-y^2}$$

can be bounded by the same mechanism *only* when $|x| < 2$ and $|y| < 3$. Under these conditions,

$$F(x,y) \geq \frac{25}{13-x^2-y^2}.$$

If in addition $xy = -1$, then $x^2 + y^2 \geq 2|xy| = 2$, so

$$\boxed{F(x,y) \geq \frac{25}{11}.}$$

However this last bound need not be sharp: equality in Cauchy would require $2/(4-x^2) = 3/(9-y^2)$, and equality in $x^2 + y^2 \geq 2$ would require $x^2 = y^2 = 1$, conditions which cannot both hold. It is an instructive valid *estimate*, not necessarily the optimum.

§1.2 A square-root comparison is Cauchy in disguise

Let $a, b, c, d, m, n > 0$. Define

$$P = \sqrt{ab} + \sqrt{cd}, \qquad Q = \sqrt{ma+nc}\sqrt{b/m + d/n}.$$

Apply Cauchy to

$$u = (\sqrt{ma}, \sqrt{nc}), \qquad v = (\sqrt{b/m}, \sqrt{d/n}).$$

[1] https://web.mit.edu/18.06/www/index.shtml

Their inner product is P, and $\|u\|\|v\| = Q$. Hence $P \le Q$. Equality occurs precisely when u and v are positive scalar multiples, equivalently

$$\frac{\sqrt{ma}}{\sqrt{b/m}} = \frac{\sqrt{nc}}{\sqrt{d/n}} \quad \Longleftrightarrow \quad \boxed{m^2ad = n^2bc}.$$

The equality condition is a useful check on the expansion of Q^2.

§1.3 Maximizing two radicals on a fixed interval

For $3 \le x \le 5$, consider

$$f(x) = 2\sqrt{x-3} + \sqrt{5-x}.$$

Cauchy–Schwarz gives

$$f(x)^2 \le (2^2 + 1^2)((x-3) + (5-x)) = 10.$$

Equality holds when

$$(\sqrt{x-3}, \sqrt{5-x}) = \lambda(2,1) \quad \text{with } \lambda \ge 0.$$

Thus $(x-3)/(5-x) = 4$, yielding $x = 23/5$. Direct substitution confirms

$$\boxed{\max_{3\le x\le 5} f(x) = \sqrt{10}.}$$

This proof establishes both a universal upper bound and a point where it is attained; there is no need to differentiate and then check endpoints separately.

§1.4 Equality in Cauchy recovers unknown vectors

Suppose $u = (a,b,c)$ and $v = (x,y,z)$ satisfy

$$\|u\|^2 = 25, \quad \|v\|^2 = 36, \quad \langle u, v\rangle = 30.$$

Cauchy gives $\langle u, v\rangle \le \|u\|\|v\| = 5 \cdot 6 = 30$. Equality forces $u = \lambda v$ for some $\lambda \ge 0$. Comparing norms gives

$$u = \frac{5}{6}v.$$

Consequently $a+b+c = \frac{5}{6}(x+y+z)$. The ratio $(a+b+c)/(x+y+z)$ equals $5/6$ *only if* the denominator is nonzero. When $x+y+z = 0$, both sums vanish, so their ratio is $0/0$ and is undefined. The proportionality of the vectors remains valid.

§1.5 Triangle angles and sharp symmetric bounds

If $A, B, C > 0$ are the angles of a Euclidean triangle, $A + B + C = \pi$. Engel's inequality gives

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} \ge \frac{(1+1+1)^2}{A+B+C} = \frac{9}{\pi}.$$

Cauchy also gives

$$A^2 + B^2 + C^2 \ge \frac{(A+B+C)^2}{3} = \frac{\pi^2}{3}.$$

The equality cases agree: both are attained by the equilateral triangle $A = B = C = \pi/3$.

§1.6 Two spheres: subtraction gives the radical plane

Let $c_1, c_2 \in \mathbb{R}^3$ and consider spheres

$$\|x - c_1\|^2 = r_1^2, \qquad \|x - c_2\|^2 = r_2^2.$$

Subtracting their equations cancels $\|x\|^2$ and leaves

$$2(c_2 - c_1) \cdot x = r_1^2 - r_2^2 + \|c_2\|^2 - \|c_1\|^2.$$

If $c_1 \neq c_2$, this is a plane, called the radical plane. Depending on how that plane meets either sphere, the intersection is a circle, a single point, or empty.

For a separate constrained optimization, maximize $w \cdot x$ on $\|x - c\| = r$. Writing $x = c + ru$, $\|u\| = 1$, Cauchy gives

$$w \cdot x = w \cdot c + r(w \cdot u) \leq w \cdot c + r\|w\|,$$

with equality at $u = w/\|w\|$ when $w \neq 0$. For $w = (-8, 6, -24)$, $\|w\| = 26$. Thus the maximum is $w \cdot c + 26r$, with its location determined explicitly by the coefficient vector.

§1.7 A product constraint and AM–GM

Suppose $a, b, c > 0$ and $(1 + a)(1 + b)(1 + c) = 8$. Since

$$(1 + a)(1 + b)(1 + c) \geq \left(1 + (abc)^{1/3}\right)^3,$$

we have $2 \geq 1 + (abc)^{1/3}$ and therefore $abc \leq 1$.

To justify the middle inequality, let $t = (abc)^{1/3}$. AM–GM gives $a + b + c \geq 3t$ and $ab + bc + ca \geq 3t^2$, so expansion of the left side is at least $1 + 3t + 3t^2 + t^3$. Equality requires $a = b = c = 1$, where $abc = 1$ and the constraint holds. The bound is therefore sharp.

§1.8 Three rays turn a radical into a triangle

Consider the inequality, valid for arbitrary real x, y, z,

$$\sqrt{x^2 + xy + y^2} + \sqrt{x^2 + xz + z^2} \geq \sqrt{y^2 + yz + z^2}.$$

Choose three unit vectors e_1, e_2, e_3 in the plane separated by $120°$, so $e_i \cdot e_j = -1/2$ for $i \neq j$. Put

$$A = xe_1, \qquad B = ye_2, \qquad C = ze_3.$$

Then

$$\|A - B\|^2 = x^2 + xy + y^2, \quad \|A - C\|^2 = x^2 + xz + z^2, \quad \|B - C\|^2 = y^2 + yz + z^2.$$

The proposed inequality is exactly $\|A - B\| + \|A - C\| \geq \|B - C\|$. It is the triangle inequality for three explicit points. The three-ray construction is the missing geometric detail: simply identifying each radical separately as a vector norm would not prove the relationship among them.

§1.9 Quadratic constraints and eigenvalues

If H is a real symmetric matrix, the extremal values of $x^T Hx$ on the unit sphere equal its smallest and largest eigenvalues. Indeed, choose an orthonormal eigenbasis, write $x = \sum_i c_i e_i$, and calculate

$$\frac{x^T Hx}{x^T x} = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2}.$$

The quotient is a weighted average of the eigenvalues and lies between their minimum and maximum. Equality occurs on the corresponding eigenspaces.

This links inequality practice to constrained optimization without replacing concrete proofs by a generic appeal to Lagrange multipliers. When a question actually contains a quadratic form and a sphere constraint, the eigenvalue calculation is its natural solution.